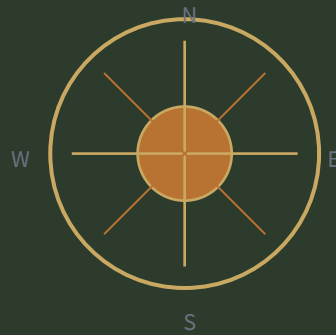


NAVIGATING THE MAZE OF COMMUNICATING ARCHITECTURE DECISIONS



Agenda

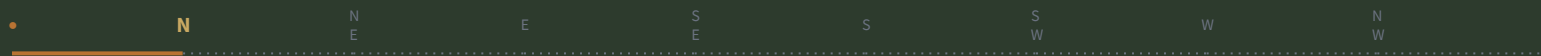
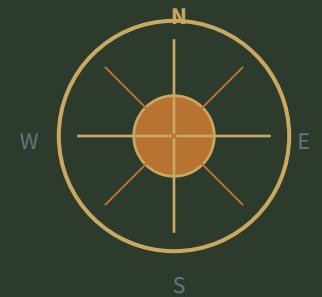
- ⦿ Introduction to ADRs
- ⦿ Core Principles
- ⦿ Decision-Making & Stakeholders
- ⦿ Best Practices
- ⦿ ADR Lifecycle
- ⦿ Real-World Examples
- ⦿ Scaling ADRs
- ⦿ Copilot Demo
- ⦿ Conclusion



Bearing: North

True North

Introduction to ADRs



What are Architecture Decision Records (ADRs)?



What are Architecture Decision Records (ADRs)?



Lightweight documents that capture these important decisions



What are Architecture Decision Records (ADRs)?



decision blueprint



What are Architecture Decision Records (ADRs)?



**Provide transparency,
accountability, and
traceability**



Historical Context

Early Foundations
(1980s-2000s):
The Pre-ADR Era



Historical Context

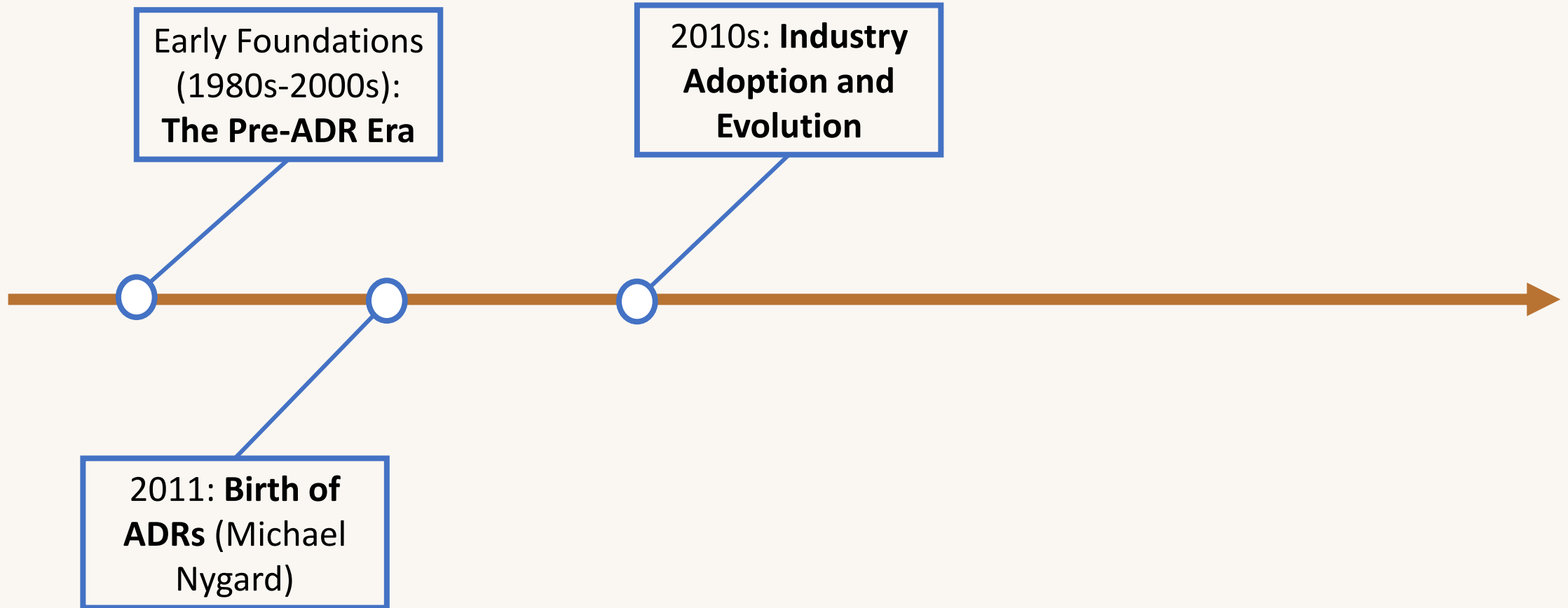
Early Foundations
(1980s-2000s):
The Pre-ADR Era



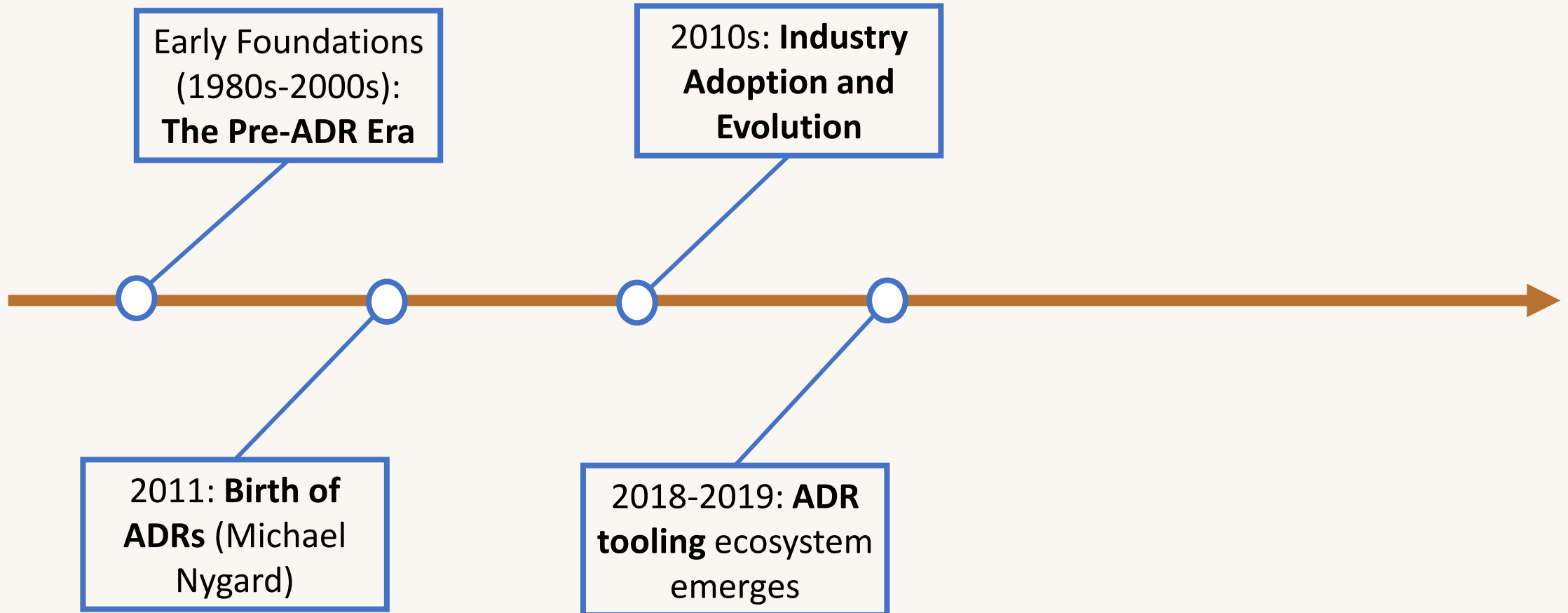
2011: **Birth of
ADRs** (Michael
Nygard)



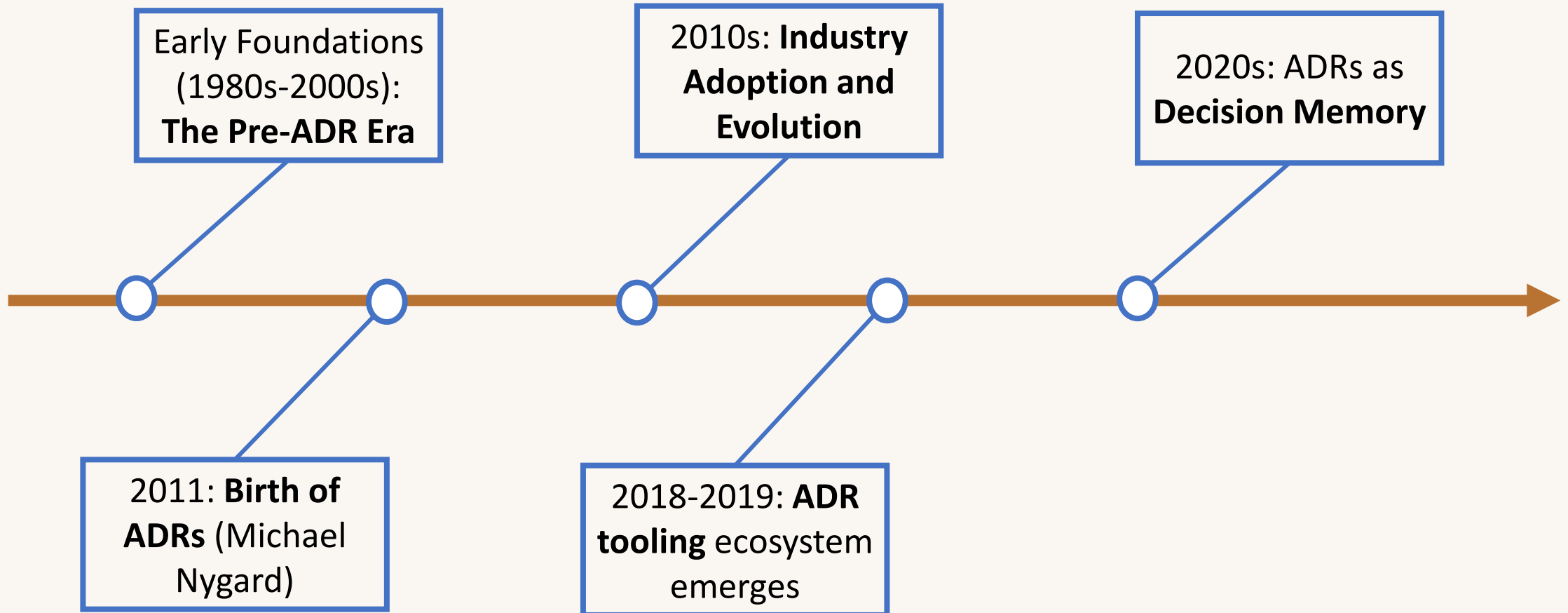
Historical Context



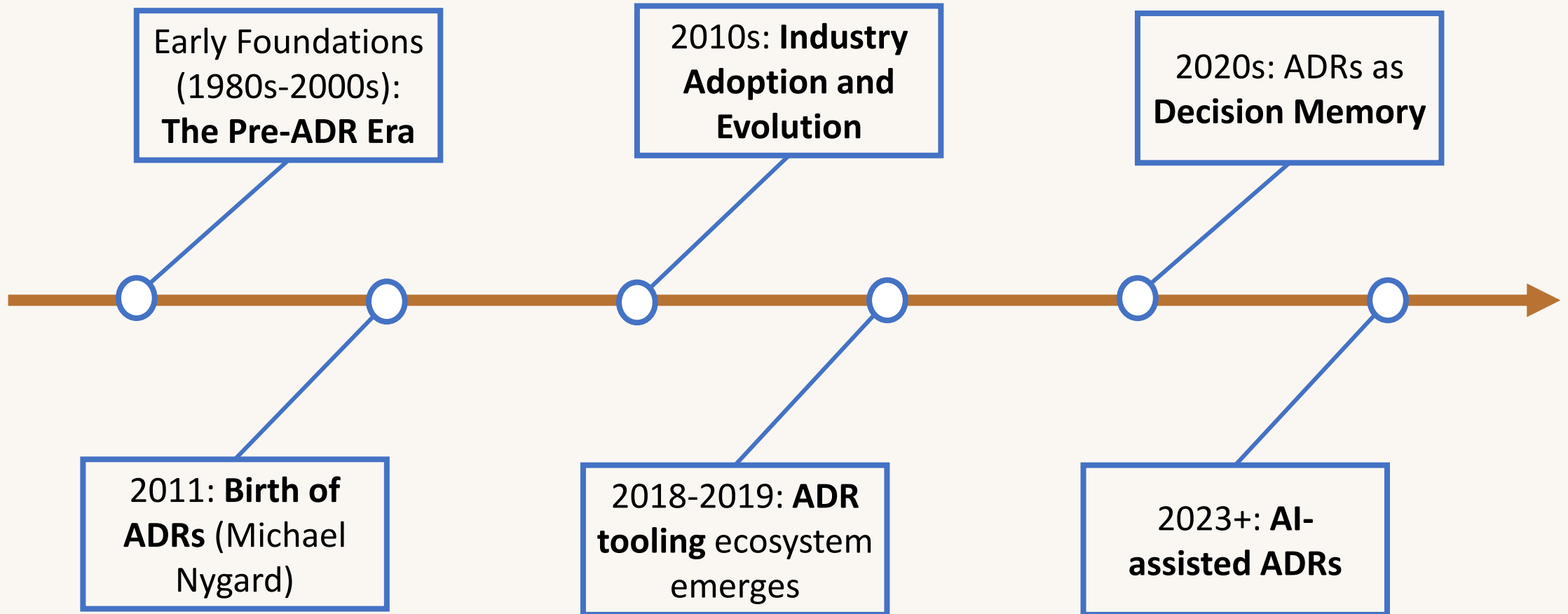
Historical Context



Historical Context



Historical Context



Why are ADRs Important?



Why are ADRs Important?



Why did we decide to do things this way?



Why are ADRs Important?



Reduce the need for back-and-forth discussions



Why are ADRs Important?



Enhance alignment across teams



Why are ADRs Important?



Promote transparency and accountability



Why are ADRs Important?



**ADRs clarify, reduce friction,
and ensure the architecture
evolves thoughtfully**



Common Misconceptions



Misconception

ADR's are only for major decisions

Reality

ADRs work best for decisions with lasting impact and trade-offs.



Common Misconceptions



Misconception

ADRs slow down development

Reality

ADRs reduce rework and facilitate faster future decisions



Common Misconceptions



Misconception

ADRs must be long and comprehensive

Reality

Effective ADRs are concise and focused



Common Misconceptions



Misconception

One ADR template fits all teams

Reality

Teams adapt templates to their specific needs



ADR Success Metrics



© **Decision Clarity**



ADR Success Metrics



© Decision Clarity

© **Onboarding Impact**



ADR Success Metrics



- © Decision Clarity
- © Onboarding Impact
- © **Decision Consistency**



ADR Success Metrics



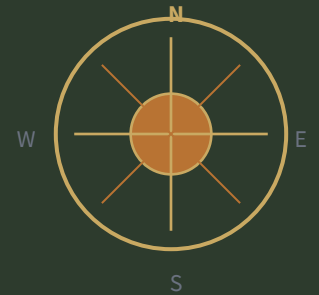
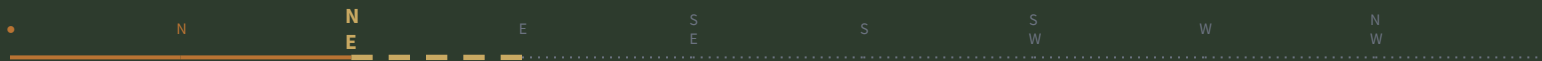
- © Decision Clarity
- © Onboarding Impact
- © Decision Consistency
- © **Rework Reduction**



Bearing: Northeast

Charting the Territory

Core Principles of ADRs



ADR Key Elements: Decision



**clear, concise statement of
the choice that was made**



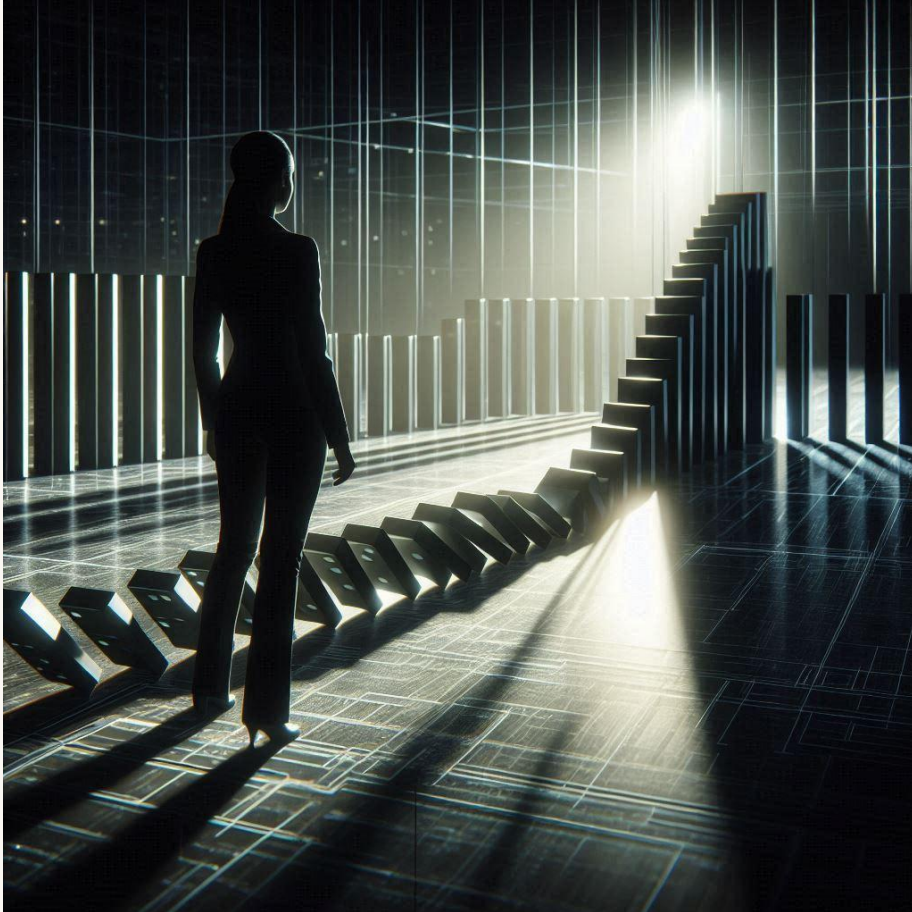
ADR Key Elements: Context



**background or problem that
led to the decision**



ADR Key Elements: Consequences



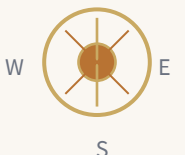
**documents the expected
outcomes of the decision**



ADR Key Elements: Status



Proposed	Decision is under consideration; not yet final
Accepted	Decision has been approved and will be implemented
Implemented	Decision has been realized in the system
Deprecated	Decision is no long valid; has been replaced
Superseded	Previous decision replaced by newer decision (with reference)



ADR Key Elements: Alternatives Considered



**documents options evaluated
and**



ADR Template

[Short title of solved problem and solution]

- **Status:** [proposed | rejected | accepted | deprecated | superseded by `xxxx`]
- **Date:** [YYYY-MM-DD when the decision was last updated]
- **Work Item:** [Identifier and name of the work item initiating the decision linked to the work item and in the form of `(xxxx-yyy)` where `xxxx` is the number of the work item and `yyy` is the name of the work item.]

Context and Problem

[Describe the context and problem in which the decision is being made in free form using a couple of sentences or in form of an illustrative story.]

<!-- Decision Drivers is an optional element; feel free to remove. -->

Decision Drivers

- [Decision driver 1, e.g., a force, facing concern, ...]
- [Decision driver 2, e.g., a force, facing concern, ...]
- ... <!-- numbers of drivers can vary -->

Considered Options

- [Title of Option 1]
- [Title of Option 2]
- [Title of Option 3]
- ... <!-- the number of options can vary -->

Decision Outcome

Chosen option: "[title of chosen option]," because [justification, e.g., the only option, which meets criterion decision driver | which resolves force (force) | ... | comes out best (see below)]

Consequences

- Good, because [positive consequence, e.g., improvement of one or more desired qualities, ...]
- Bad, because [negative consequence, e.g., compromising one or more desired qualities, ...]

Implementation

[Describe the steps required to implement the decision, including any dependencies, timelines, or resource allocations.]

Confirmation

[Describe how the implementation of/compliance with the ADR is confirmed.]

Stakeholders

[List the stakeholders involved in or affected by the decision. This may include individuals, teams, departments, or external parties.]

Pros and Cons of the Options

[Title of Option 1]

- Good, because [argument a]
- Good, because [argument b]
- Neutral, because [argument c]
- Bad, because [argument d]
- ... {numbers of pros and cons can vary}

[Title of Other Option]

- Good, because [argument a]
- Good, because [argument b]
- Neutral, because [argument c]
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More Information

[You might want to provide additional evidence/confidence for the decision outcome here and/or document the team agreement or the decision and/or define when/how this decision should be realized and if/when it should be re-visited. Links to other decisions and resources might appear here as well. If there is no more information, please use "No more information available".]

Follow-On Information

[Provide details about what happens to the ADR after approval or rejection (and the record is immutable). This section can include information about any updates, revisions, or further actions taken based on the decision outcome.]

Record History

- **Proposed:** [Date]
- **Rejected:** [Date, if applicable]
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- **Date Superseded:** [Date, if applicable]
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ADR Template: Title and Metadata

[Short title of solved problem and solution]

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ADR Template: Context and Problem

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ADR Template: Decision Drivers

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ADR Template: Considered Options

Considered Options

- [Title of Option 1]
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ADR Template: Decision Outcome

Decision Outcome

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ADR Template: Pros and Cons

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ADR Template: More Information

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ADR Template: Follow-On Information

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- **Date Superseded:** [Date, if applicable]
- **Archived:** [Date, if applicable]



Template Variations

Minimalist

- Status
- Context
- Decision
- Consequences
- Rational

Standard

- Status
- Context
- Decision
- Consequences
- Alternatives

Technical Deep-Dive

- | | | |
|---|---|--|
| <ul style="list-style-type: none">• Status• Context• Technical Requirements | <ul style="list-style-type: none">• Decision• Architecture Diagrams• Alternative with Performance Metrics | <ul style="list-style-type: none">• Implementation Details• Migration Path• Consequences (Technical + Operational) |
|---|---|--|



Writing Clear ADRs



Writing Clear ADRs



© Know Your Audience



Writing Clear ADRs



© Know Your Audience

© **Be concise but comprehensive**



Writing Clear ADRs



- © Know Your Audience
- © Be concise but comprehensive
- © **Focus on the “why”, not just the what**



Writing Clear ADRs



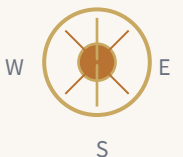
- © Know Your Audience
- © Be concise but comprehensive
- © Focus on the “why”, not just the what
- © **Address alternatives and rejected options**



Writing Clear ADRs



- © Be concise but comprehensive
- © Focus on the “why”, not just the what
- © Address alternatives and rejected options
- © **Keep the language simple and direct**



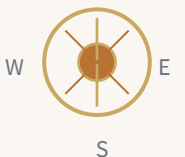
Writing Clear ADRs

Vague ADR

We chose MongoDB.

More Clarity

We chose MongoDB for its flexible schema and high write throughput, which addresses our scalability needs.



Anti-Examples Gallery

**Empty
Consequences**

**Strawman
Alternatives**

**Never-Updated
ADR**

Context Blackhole

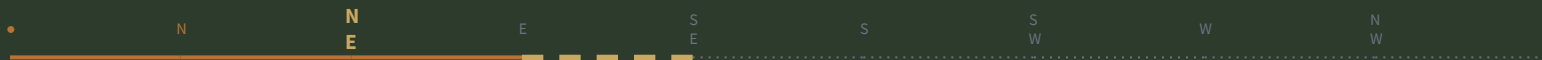
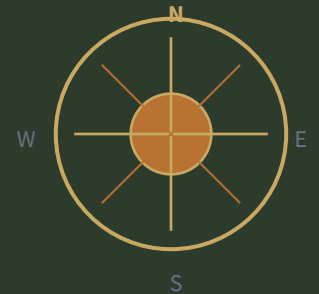
**Never-Updated
ADR**



Bearing: East

Gathering the Expedition

Decision Making & Stakeholders



Who Should be Involved?

Architects

**Team Lead / Tech
Lead**

**Operations /
DevOps**

**Product
Management**

**Security /
Infrastructure**



Stakeholder Matrix

Stakeholder	R	A	C	I
Architect	X	✓		✓
Tech Lead	✓		X	✓
Operations			✓	✓
Product			X	✓
Security			X	✓

R = Responsible: Does the work / makes the recommendation

A = Accountable: Final decision authority

C = Consulted: Provides input before the decision

I = Informed: Kept in loop; receives communication



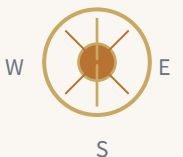
Gathering Input

Synchronous Methods

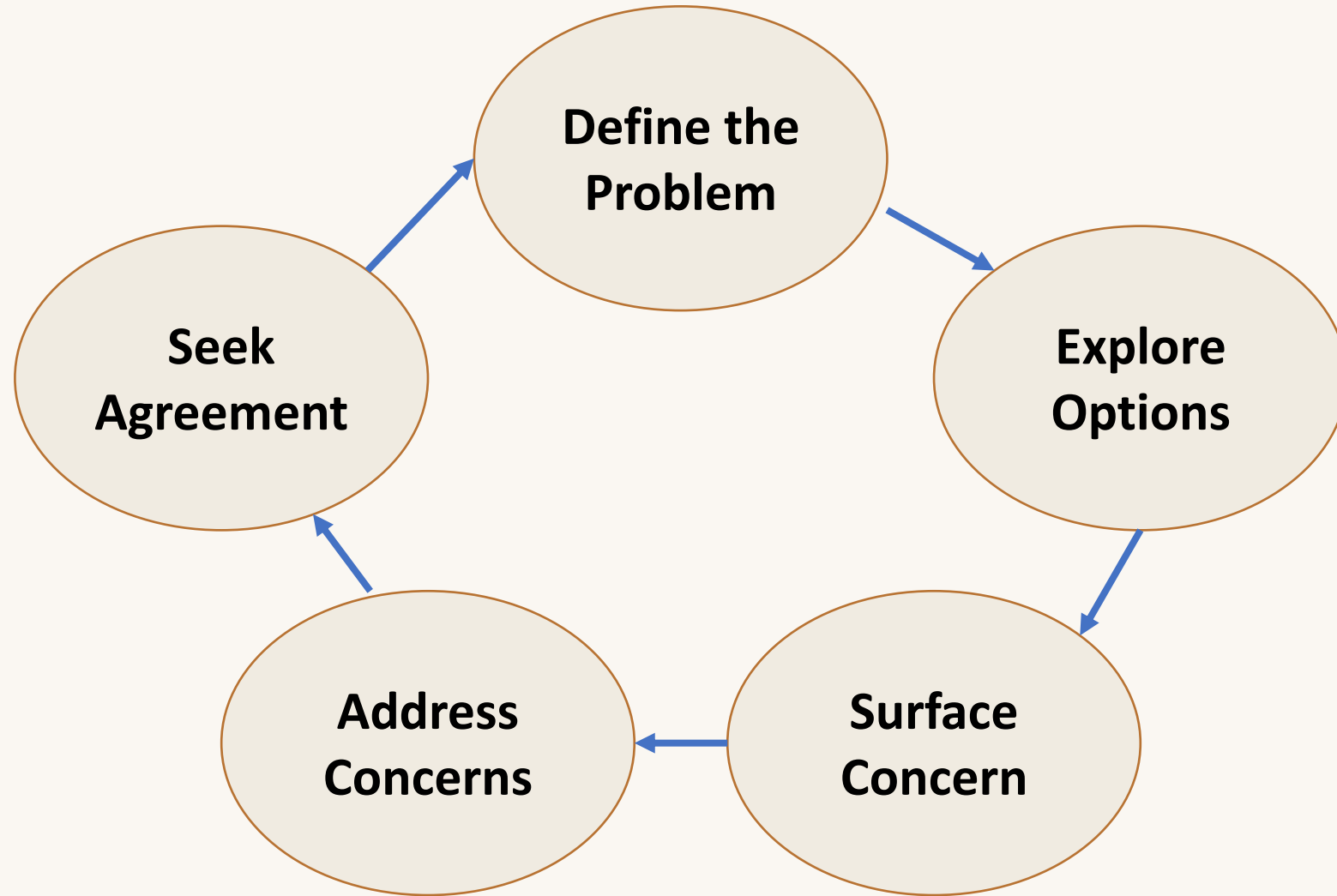
- Architecture Review Meetings
- Design Review Workshops
- Ad-Hoc Hallway Conversations

Asynchronous Methods

- RFC Documents (Request for Comments)
- Email / Discussion Forum
- Decision Documentation Drafts
- Teams/Slack/Chat Channels



Building Consensus



When Consensus Fails

Escalate to Higher Authority

Disagree and Commit

Record Dissent



Decision Authority

Architect Authority

Tech Lead Authority

Consensus Vote

Leadership Override



Stakeholder Communication



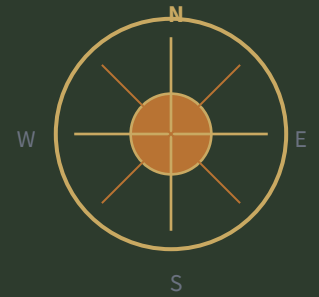
- © Decision Announcement (All Stakeholders)
- © Detailed Explanation (Involved Stakeholders)
- © Implementation Guidance (Engineering Team)
- © Ongoing Transparency (All)



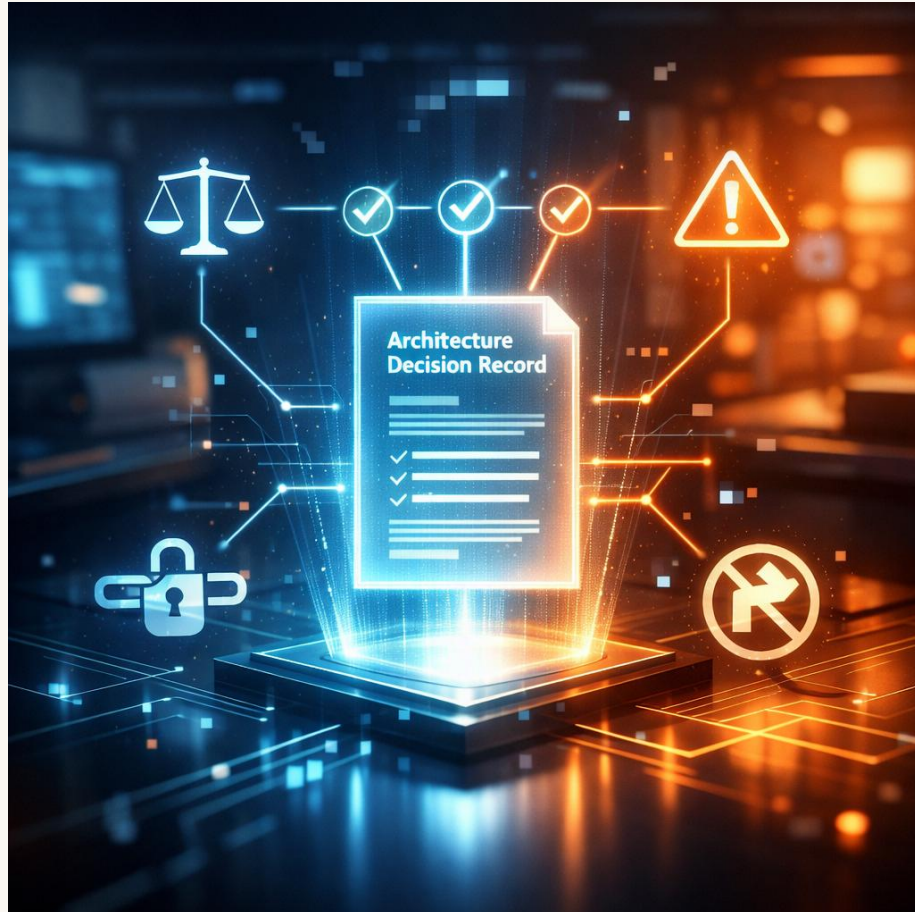
Bearing: Southeast

Refinement

Best Practices



When to Write an ADR

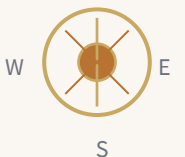


Write an ADR if:

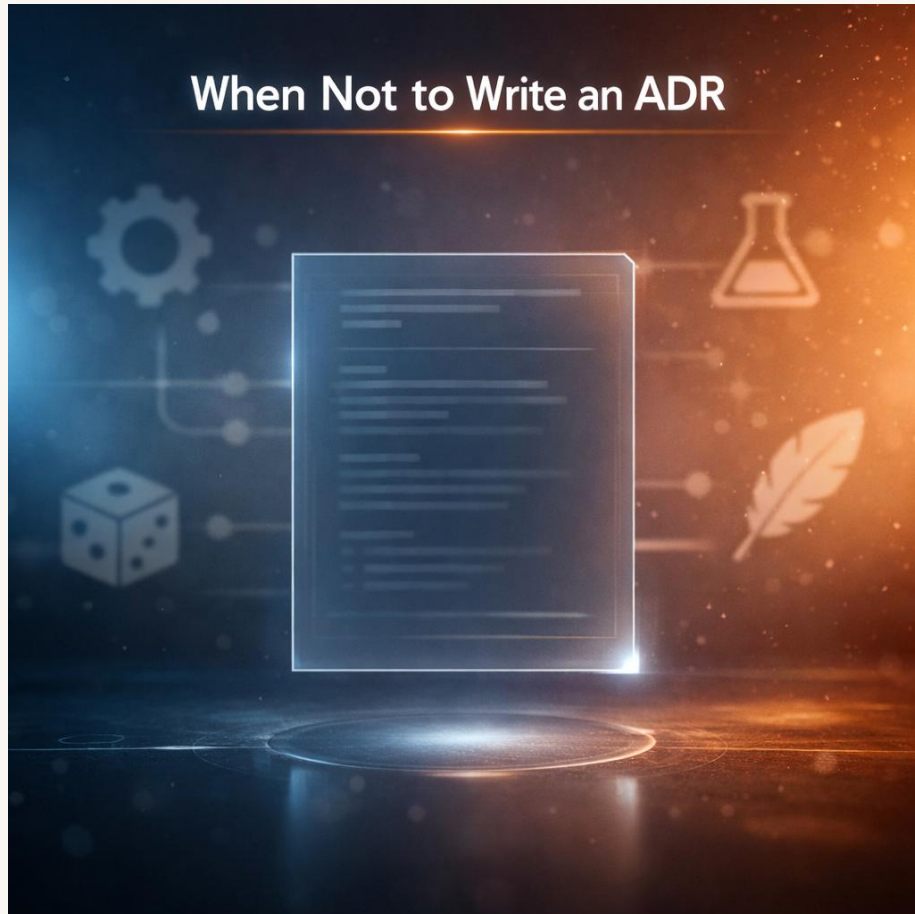
- Decision impacts multiple teams or systems
- Decision is difficult to reverse
- Decision shapes architecture for months/years
- Decision involves significant trade-offs
- Decision impacts operational complexity
- Future team members need context on “why”
- Decision required cross-functional discussion

Document Informally if:

- Low-impact local optimization
- Trivial/obvious decision
- Reversible with minimal cost
- Team already has shared context



When NOT to Write an ADR



- © Micro-Decisions
- © Solution Rather Than Decision
- © Documentation by Committee



Storage and Organization

Git Repository

Central Wiki/Portal

Hybrid



ADR Numbering Schemes

Sequential

0001, 0002, 0003...

Date-Based

YYYY-MM-DD-descriptor

Hierarchical

Team.Domain.Number

Semantic

Decision type indicator



Integration with Documentation Portals



- ◎ Manual Integration
- ◎ Semi-Automated (Index + Links)
- ◎ Fully Automated (Portal Feeds)



Review and Approval Process

Draft (Author)

Stakeholder Review (Reviewers)

Revision (Author)

Approval Decision (Decision Authority)

Acceptance & Communication



Keeping ADRs Discoverable



- © Search & Indexing
- © Navigation & Taxonomy
- © Relationship & Cross-Linking
- © Regular Review & Maintenance



ADR Quality Checklist

Content

- ☐ Decision is clearly stated (can be extracted in one sentence)
- ☐ Context explains why this decision was necessary
- ☐ Consequences are specific and foreseeable (not vague)
- ☐ Alternatives considered are substantive (not strawmen)
- ☐ Tradeoffs are explicit ("we gain X, we lose Y")
- ☐ Status is clearly indicated (Proposed, Accepted, Deprecated, Superseded)



ADR Quality Checklist

Content

Clarity

- ☐ Written for future developers (not just current team)
- ☐ Technical depth appropriate for decision scope
- ☐ Jargon is minimal or explained
- ☐ Examples provided were helpful
- ☐ Formatting is readable (headings, lists, whitespace)



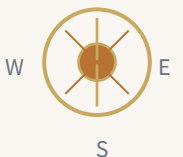
ADR Quality Checklist

Content

Clarity

Process

- ☐ Key stakeholders were consulted
- ☐ Dissenting opinions (if any) are documented
- ☐ Decision authority is identified and has approved
- ☐ Rationale for decision authority's role is clear



Measuring ADR Effectiveness

Usage

- % of architecture decisions documented as ADRs (Target 80%)
- # of ADRs created per quarter (track trend)
- # of ADR references in code reviews (show integration with dev)
- # of ADRs superseded/deprecated (show health evolution)



Measuring ADR Effectiveness

Usage

Quality

- Average # of stakeholders consulted per ADR
- % of ADRs with documented alternatives considered
- % of ADRs with consequences section
- Average time from proposed to acceptance



Measuring ADR Effectiveness

Usage

Quality

Outcome

- Time to onboard new engineers, explaining architectural choices
- # of “why did we do this?” questions in code review
- Team confidence in architecture understanding
- Time to reverse failed decisions (after documented rationale review)



Measuring ADR Effectiveness

Usage

Quality

Outcome

Cost

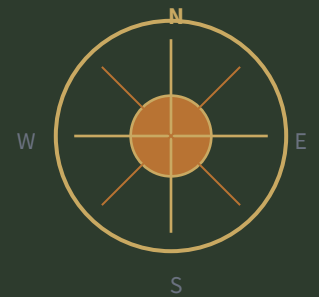
- Time invested in writing/reviewing ADRs per quarter
- Comparison to time wasted in re-deciding or reverse decisions
- ROI: hours saved in onboarding/hours invested in ADRs



Bearing: Southwest

Landmarks & Lessons

Real-World Examples



The Decision



**Adopt a serverless
architecture using Azure
Functions**



Rationale



**reduced need for
infrastructure management**

**Azure Functions provided
automatic scaling**



Context



needed to move fast and stay competitive in a fast-evolving market



Consequences

Positives

- © Faster time-to-market
- © Reduced operational costs
- © Improved scalability



Consequences

Positives

- © Faster time-to-market
- © Reduced operational costs
- © Improved scalability

Trade-Offs

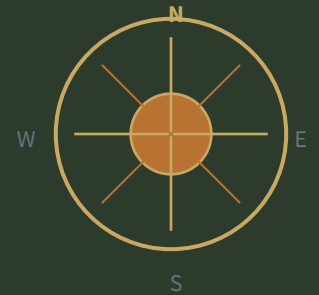
- © Initial learning curve
- © Vendor lock-in



Bearing: Northwest

The Digital Compass ✨

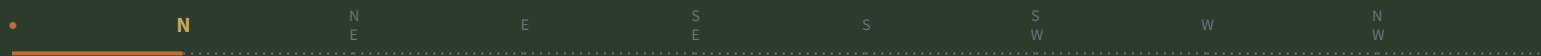
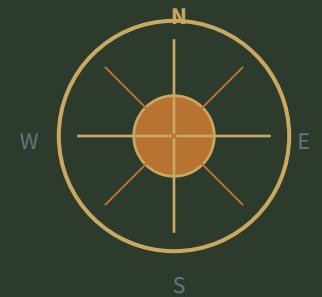
AI Demo



Bearing: North

Full Circle

Conclusion



Key Takeaways



- ADRs Capture the “Why” Behind Technical Decisions
- Inclusive Decision-Making Creates Better Architecture
- Decision Governance Must Evolve as Your Scale
- ADRs Are Living Documents, Not Archives
- Implementation Requires Culture Adoption, Not Just Process



The Implementation Roadmap



Foundation



The Implementation Roadmap



Foundation

Adoption



The Implementation Roadmap



Foundation

Adoption

Refinement



The Implementation Roadmap

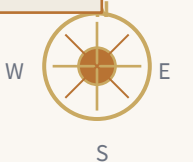


Foundation

Adoption

Refinement

Institutionalization



Common Pitfalls to Avoid

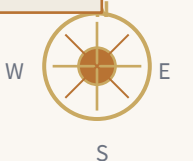


Over-Documentation

No Review Process

Lost in the Archive

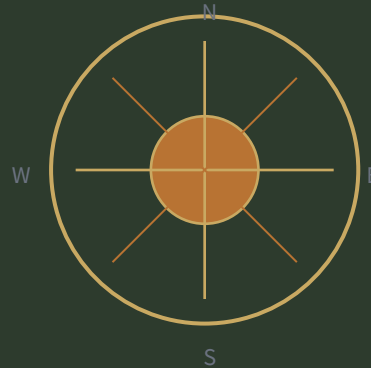
Committee Decision-Making



Q&A



Journey's End



Start capturing the "why" behind your decisions today.